A Timeline of Artificial Intelligence

From the 1940s to generative AI, with two AI winters along the way

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Eight decades in seven eras



Red eras were periods of progress and hype. **Gray** eras were AI winters, when funding and interest collapsed.

Foundations: from theory to a new field

1943

McCulloch and Pitts model an artificial neuron

1950

Turing proposes the imitation game, now called the Turing Test

1956

Dartmouth Workshop coins “artificial intelligence”

1958

Rosenblatt demonstrates the Perceptron, an early trainable network

Early optimism, then the first AI winter

1966 ELIZA, an early chatbot, imitates a therapist with pattern matching

1969 Minsky and Papert show the limits of single-layer networks

1973 The Lighthill Report and DARPA cuts shrink funding

First AI winter, 1973–1980

Causes

- Promises went unmet
- Computing power and memory were limited

Impacts

- Funding cut in the US and UK
- Labs shrank; researchers moved on

Expert systems boom and the second winter

1980s Expert systems such as XCON bring commercial success

1986 Backpropagation is popularized for multi-layer networks

1987 The Lisp machine market collapses

Second AI winter, 1987–1993

Causes

- Specialized hardware lost to cheaper PCs
- Expert systems were costly and brittle

Impacts

- Companies failed and investment dried up
- Many avoided the “AI” label

Machines beat human champions

1997

Deep Blue

IBM’s chess computer defeats world champion Garry Kasparov

2011

Watson

IBM Watson wins Jeopardy! against human champions

2016

AlphaGo

DeepMind’s system defeats Go champion Lee Sedol

Why deep learning finally took off

1. **GPUs**
   Nvidia hardware made large networks practical; AlexNet won ImageNet in 2012

2. **Cloud computing**
   AWS launched in 2006; Azure and Google Cloud followed

3. **Data**
   The internet and datasets like ImageNet supplied training data

4. **Algorithms**
   Backpropagation, convolutional networks, and the 2017 Transformer

Key figures: Geoffrey Hinton, Yann LeCun, Yoshua Bengio, Ian Goodfellow, Demis Hassabis, Ilya Sutskever, Fei-Fei Li

The generative AI era

2014

GANs introduced by Ian Goodfellow

2017

Transformer architecture from Google

2018–20

BERT, GPT-2, and GPT-3 show the power of scale

2022

ChatGPT launches and reaches mass adoption

2023–24

GPT-4, Claude, Gemini, Llama; AI work wins Nobel Prizes

Leading AI companies and models

Company	Notable models	Role
OpenAI	GPT series, ChatGPT	Started the 2022 consumer wave
Google DeepMind	Gemini, AlphaGo, AlphaFold	Invented the Transformer
Anthropic	Claude	Safety-focused AI company
Meta	Llama	Open-weight models
Microsoft	Copilot, Azure AI	Cloud provider, OpenAI partner
Nvidia	GPUs, CUDA	Hardware for AI training

What the timeline teaches

Hype cycles repeat

Bold promises without results led to both AI winters.

Infrastructure matters

GPUs, cloud, and data mattered as much as new algorithms.

People stay responsible

Tools get more capable, but people choose how to use them.

References

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